# Worksheet 4.5 — Conditional Probability and Tree Diagrams

Mathematics: Analysis & Approaches SL (IB) · Unit 4: Statistics and Probability

### The Big Ideas

Tree diagrams handle sequences of events, including without-replacement draws and reversed conditionals.

- $P(A \mid B) = \dfrac{P(A \cap B)}{P(B)}$.
- Multiply along a branch; add across matching outcomes.

## Worked Examples

### Example 1 — Basic conditional probability

$P(A) = 0.6$, $P(A \cap B) = 0.18$. Find $P(B \mid A)$.

***Solution.*** $P(B \mid A) = \dfrac{0.18}{0.6}$.

**Conclusion:** 0.3.

### Example 2 — Without replacement

A bag has 6 red, 4 blue balls. Two drawn without replacement; find $P(\text{both red})$.

***Solution.*** $\dfrac{6}{10} \times \dfrac{5}{9} = \dfrac{30}{90}$.

**Conclusion:** $\dfrac{1}{3}$.

### Example 3 — At least one

Using the same bag, find $P(\text{at least one blue})$.

***Solution.*** $1 - P(\text{both red}) = 1 - \frac{1}{3}$.

**Conclusion:** $\dfrac{2}{3}$.

### Example 4 — Reverse conditional (tree reversal)

Factory A makes 65% of items with a 2% defect rate; Factory B makes 35% with a 6% defect rate. Given an item is defective, find $P(\text{from B})$.

***Solution.*** $P(A \cap d) = 0.65(0.02) = 0.013$; $P(B \cap d) = 0.35(0.06) = 0.021$; total $= 0.034$.

**Conclusion:** $P(B \mid d) = \dfrac{0.021}{0.034} \approx 0.618$.

### Example 5 — Three-stage without-replacement sequence

A bag has 5 red, 3 blue balls. Three drawn without replacement; find $P(\text{all three red})$.

***Solution.*** $\frac{5}{8} \times \frac{4}{7} \times \frac{3}{6} = \frac{60}{336}$.

**Conclusion:** $\frac{60}{336} = \frac{5}{28}$.

### Example 6 — Conditional probability from a two-way table

Of 100 students, 60 play sports; of those, 40 also do music. Of the 40 who don't play sports, 10 do music. Find $P(\text{music} \mid \text{sports})$.

***Solution.*** $P(\text{music} \mid \text{sports}) = \frac{40}{60}$.

**Conclusion:** $\frac{2}{3}$.

### Example 7 — Comparing conditional probabilities both ways

Using the same data (100 students; 60 sports, 40 of them music; 40 non-sports, 10 of them music), find $P(\text{sports} \mid \text{music})$ and compare to $P(\text{music} \mid \text{sports})$.

***Solution.*** Total music $= 40 + 10 = 50$. $P(\text{sports} \mid \text{music}) = \frac{40}{50} = 0.8$.

**Conclusion:** $P(\text{sports} \mid \text{music}) = 0.8 \neq P(\text{music} \mid \text{sports}) = \frac{2}{3} \approx 0.667$ — confirming order matters.

### Example 8 — Independence test using a tree diagram's branches

A spinner has $P(\text{red}) = 0.4$ on each of two independent spins. Find $P(\text{exactly one red})$ using a tree diagram.

***Solution.*** Two paths: (red, not red) and (not red, red), each with probability $0.4 \times 0.6 = 0.24$.

**Conclusion:** $P(\text{exactly one red}) = 0.24 + 0.24 = 0.48$.

### Example 9 — Application — medical test reliability (reverse conditional)

A disease affects 2% of a population. A test correctly detects it 95% of the time (true positive) and gives a false positive 3% of the time. Given a positive result, find $P(\text{has the disease})$.

***Solution.*** $P(\text{disease} \cap +) = 0.02(0.95) = 0.019$. $P(\text{no disease} \cap +) = 0.98(0.03) = 0.0294$. Total $P(+) = 0.0484$.

**Conclusion:** $P(\text{disease} \mid +) = \frac{0.019}{0.0484} \approx 0.393$ — surprisingly low, since the disease is rare.

## Your Turn — Show Your Work

For each question, show all steps clearly in the space provided.

### Question 1

$P(A) = 0.7$, $P(A \cap B) = 0.28$. Find $P(B \mid A)$.

### Question 2

A bag has 5 green, 5 yellow marbles. Two drawn without replacement; find $P(\text{both green})$.

### Question 3

Using the same bag, find $P(\text{at least one yellow})$.

### Question 4

Two independent events each have probability 0.3. Find $P(\text{exactly one occurs})$.

### Question 5 — Challenge

A box has 9 working, 3 faulty phones. Two selected without replacement; find $P(\text{at least one faulty})$.

## Question 6

Machine X makes 40% of output with 5% defects; Machine Y makes 60% with 2% defects. Find the overall $P(\text{defective})$.

## Question 7 — Challenge

Using the previous question, given an item is defective, find $P(\text{from Machine X})$.

## Question 8

A bag has 6 red, 4 blue, 2 green balls. Three drawn without replacement; find $P(\text{all three red})$.

## Question 9

Of 120 students, 70 play an instrument; of those, 30 also play a sport. Of the 50 who don't play an instrument, 20 play a sport. Find $P(\text{sport} \mid \text{instrument})$.

### Question 10

Using the previous question's data, find $P(\text{instrument} \mid \text{sport})$ and state whether it equals $P(\text{sport} \mid \text{instrument})$.

### Question 11 — Challenge

A spinner has $P(\text{blue}) = 0.3$ on each of two independent spins; find $P(\text{exactly one blue})$.

### Question 12 — Challenge

A disease affects 1% of a population; a test detects it correctly 90% of the time and gives a false positive 4% of the time. Given a positive result, find $P(\text{has the disease})$.

### Question 13 — Challenge

A box has 10 working, 2 faulty phones. Two selected without replacement; find $P(\text{exactly one faulty})$.

# Question Recap & Answer Key

## Question Recap

1. $P(A) = 0.7$, $P(A \cap B) = 0.28$. Find $P(B \mid A)$.
2. A bag has 5 green, 5 yellow marbles. Two drawn without replacement; find $P$(both green).
3. Using the same bag, find $P$(at least one yellow).
4. Two independent events each have probability 0.3. Find $P$(exactly one occurs).
5. A box has 9 working, 3 faulty phones. Two selected without replacement; find $P$(at least one faulty).
6. Machine X makes 40% of output with 5% defects; Machine Y makes 60% with 2% defects. Find the overall $P$(defective).
7. Using the previous question, given an item is defective, find $P$(from Machine X).
8. A bag has 6 red, 4 blue, 2 green balls. Three drawn without replacement; find $P$(all three red).
9. Of 120 students, 70 play an instrument; of those, 30 also play a sport. Of the 50 who don't play an instrument, 20 play a sport. Find $P$(sport | instrument).
10. Using the previous question's data, find $P$(instrument | sport) and state whether it equals $P$(sport | instrument).
11. A spinner has $P$(blue) = 0.3 on each of two independent spins; find $P$(exactly one blue).
12. A disease affects 1% of a population; a test detects it correctly 90% of the time and gives a false positive 4% of the time. Given a positive result, find $P$(has the disease).
13. A box has 10 working, 2 faulty phones. Two selected without replacement; find $P$(exactly one faulty).

## ✓ Answer Key

1. $0.28/0.7 = 0.4$
2. $\frac{5}{10} \times \frac{4}{9} = \frac{20}{90} = \frac{2}{9}$
3. $1 - \frac{2}{9} = \frac{7}{9}$
4. $2(0.3 \times 0.7) = 0.42$
5. $P$(no faulty) $= \frac{9}{12} \times \frac{8}{11} = \frac{72}{132} = \frac{6}{11}$; at least one $= 1 - \frac{6}{11} = \frac{5}{11}$
6. $0.4(0.05) + 0.6(0.02) = 0.02 + 0.012 = 0.032$
7. $P(X \mid d) = \dfrac{0.02}{0.032} = 0.625$
8. $\frac{6}{12} \times \frac{5}{11} \times \frac{4}{10} = \frac{120}{1320} = \frac{1}{11}$
9. $P$(sport | instrument) $= \frac{30}{70} = \frac{3}{7} \approx 0.429$
10. total sport players $= 30 + 20 = 50$; $P$(instrument | sport) $= \frac{30}{50} = 0.6$; not equal to $\frac{3}{7} \approx 0.429$, confirming order matters
11. $2(0.3 \times 0.7) = 0.42$
12. $P(d \cap +) = 0.01(0.9) = 0.009$; $P$(no $d \cap +$) $= 0.99(0.04) = 0.0396$; total $= 0.0486$; $P(d \mid +) = 0.009/0.0486 \approx 0.185$
13. $P$(no faulty) $= \frac{10}{12} \times \frac{9}{11} = \frac{90}{132} = \frac{15}{22}$; $P$(both faulty) $= \frac{2}{12} \times \frac{1}{11} = \frac{2}{132} = \frac{1}{66}$; exactly one $= 1 - \frac{15}{22} - \frac{1}{66} = 1 - \frac{45}{66} - \frac{1}{66} = \frac{20}{66} = \frac{10}{33}$